

TECHNICAL DATA SHEET (TDS) – PRODUCT 4 (REVISED)

PRODUCT NAME: Electrochromic Auto-Dimming Side Mirror Glass Pair

APPLICATION: BMW E38 7-Series / E39 5-Series Premium Mirrors

ORIENTATION: Left Hand (LH) & Right Hand (RH) Symmetric Set

OEM LH CROSS-REFERENCE: 51168266493

OEM RH CROSS-REFERENCE: 51168266494

1. DIMENSIONS & STRUCTURAL GEOMETRY

- **Glass Assembly Profile:** Dual-pane laminated glass structure with an internal hermetic gel cavity. Total thickness: 4.20 mm.
- **Curvature Profile:** Aspherical multi-curvature layout with integrated vertical laser-etched blind-spot separation line.
- **Backing Elements:** High-impact ABS carrier plate with integrated PTC heating element grid and standard OEM 2-pin spade connectors, and two standard lug connections for heating..

2. MATERIAL SPECIFICATIONS

- **Reflective Layer:** Vacuum-sputtered, high-reflectivity **Chromium (Cr) or Silver (Ag)** backing layer complying with **ISO 9050** (>65% initial reflectivity).
- **Active Layer:** High-stability, non-bleeding **Viologen-based Electrochromic Liquid Gel**.
- **Electrical Compatibility:** Operating input window: **1.2V to 1.5V DC** (Perfect drop-in sync with the OEM interior rear-view mirror photodiode sensor logic).

3. EXTRAVAGANT CLIMATIC SPECIFICATIONS (Anti-Bleeding Control)

- **Environmental Endurance:** Must survive **High-Temperature & High-Humidity Storage (85°C / 85% Relative Humidity for 5000 continuous cycles)** without any edge-browning, gel decomposition, delamination, or bubble formation.
 - **Dimming Kinetics:** Active transmission drop from fully clear to max tint within **< 3.0 seconds** at 30°C.
-

